

Part 4: System password hashes

Ethical and authorized use: use only `../data/linux_hashes.txt`. The parser does not access `/etc/shadow`. Never obtain or test another person's hashes.

Run `python3 analyze_shadow_hash.py ../data/linux_hashes.txt`. A shadow-style value uses `$identifier$salt-or-parameters$verifier`; the identifier selects the password-hashing scheme. Classify this exercise as online or offline and justify your answer.

Parse the supplied record

```
cd /workspace/labs/lab01/part4_system_hashes
python3 analyze_shadow_hash.py ../data/linux_hashes.txt
```

Expected output has this structure:

```
{'account': 'student_...', 'id': '6', 'algorithm': 'SHA-512-crypt',
 'salt_or_parameters': '...', 'verifier': '...'}
```

The `$6$` identifier selects SHA-512-crypt. The salt and verifier are stored data; neither field is the plaintext password. Notice that the script accepts an explicit local filename and never opens `/etc/shadow` automatically.

Optional local cracking command:

```
john --wordlist=../data/lab01-small.txt ../data/linux_hashes.txt
```

To display John the Ripper's result for this supplied file, run:

```
john --show ../data/linux_hashes.txt
```

Runtime and exact status messages can vary. The important observation is that all guesses are checked locally, without sending login attempts to a service.

Use only the supplied synthetic file. If an instructor separately provides a Victim Ubuntu VM, it must be deliberately vulnerable and connected only to an isolated host-only network. Worksheet (manual observation only): list the VM's instructor-provided address, visible services, evidence of isolation, and which service would authenticate users. Do not scan arbitrary addresses.

Checkpoint questions

1. Does the hash format reveal the plaintext?
2. Which fields must be stored so legitimate verification remains possible?
3. Is this an online or offline activity? What evidence supports your answer?
4. Why must an optional Victim VM use an isolated host-only network?